QUESTION 1

What is the primary role of a data sheet in electronics?

Answer is Correct **Score: 1 / 1**

Options

A. To provide marketing information

B. To describe the external appearance of a component

C. To explain programming logic

Correct

D. To describe an electronic component in sufficient detail for practical use

FEEDBACK

Solution: A data sheet helps users understand a component's characteristics to use it effectively in systems.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=34

QUESTION 2

What is a key distinction between a data brief and a data sheet?

Not Attempted **Score: 0 / 1**

Options

A. Data sheets include pricing

B. Data briefs are for internal testing

Expected

C. Data briefs provide high-level summaries, not full technical detail

D. Data sheets focus on software features

FEEDBACK

Solution: Data briefs offer an overview, while data sheets go into full technical specifications.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=104

QUESTION 3

Why is the frequency 32.768 kHz commonly used for oscillators in microcontrollers?

Not Attempted **Score: 0 / 1**

Options

A. It is compatible with USB protocols

B. It is the clock speed of all CPUs

Expected

C. It is 2^15 and easily divisible to get 1 Hz

D. It minimizes power consumption

FEEDBACK

Solution: 32.768 kHz = 2^15, making it easy to divide down to 1 Hz for timekeeping.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=205

QUESTION 4

Which information is typically found on the first page of a microcontroller data sheet?

Not Attempted **Score: 0 / 1**

Options

A. Assembly instructions

B. Programming examples

Expected

C. Core type, voltage range, and major peripherals

D. Price and availability

FEEDBACK

Solution: The first page usually summarizes the core architecture, voltage range, memory sizes, and key interfaces.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=524

QUESTION 5

Which feature of the STM32C031 allows it to safely interface with 5V logic devices?

Not Attempted **Score: 0 / 1**

Options

A. It has a dedicated 5V power regulator

B. It uses an external level shifter

Expected

C. Its GPIOs are 5V tolerant

D. It has optical isolation

FEEDBACK

Solution: The STM32C031's GPIO pins are 5V tolerant, allowing safe connection to 5V peripherals.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=644

QUESTION 6

What can the timers in the STM32C031 be used for, according to the data sheet?

Not Attempted **Score: 0 / 1**

Options

A. Analog to digital conversion

B. Power management

Expected

C. Pulse-width modulation (PWM)

D. EEPROM access

FEEDBACK

Solution: The timers can generate PWM signals for applications like motor control.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=90

QUESTION 7

What type of packaging is typically used for modern microcontrollers like STM32C031?

Not Attempted **Score: 0 / 1**

Options

A. Through-hole DIP only

Expected

B. Surface-mount technology

C. Socketed modules

D. Magnetic shielding

FEEDBACK

Solution: Surface-mount technology (e.g., LQFP, TSSOP) is the most common for microcontrollers.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=395

QUESTION 8

What should be used to learn about memory mapping and control registers for STM32 MCUs?

Not Attempted **Score: 0 / 1**

Options

A. Device manager

B. Compiler documentation

Expected

C. Reference manual

D. Bootloader firmware

FEEDBACK

Solution: Control registers and memory maps are detailed in the reference manual, not the data sheet.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=789

QUESTION 9

What is the safe continuous forward current for the standard red LED in the given data sheet?

Not Attempted **Score: 0 / 1**

Options

A. 5 mA

B. 10 mA

Expected

C. 20 mA

D. 100 mA

FEEDBACK

Solution: The safe continuous forward current is specified as 20 milliamps.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=1138

QUESTION 10

Why should caution be taken with reverse voltage in LEDs?

Not Attempted **Score: 0 / 1**

Options

A. It will reduce brightness

Expected

B. It can permanently damage the LED

C. It will increase temperature

D. It will disable PWM

FEEDBACK

Solution: Reverse voltages above 5V can damage the LED.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=1157

QUESTION 11

What does the LED wavelength range of 600–650 nm indicate?

Not Attempted **Score: 0 / 1**

Options

A. It emits green light

B. It is UV protected

Expected

C. It is a red LED

D. It operates at high voltage

FEEDBACK

Solution: A wavelength range of 600–650 nm corresponds to the red region of the visible spectrum.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=1256

QUESTION 12

Why is knowing physical tolerances (like ±0.25 mm) in a component's data sheet important?

Not Attempted **Score: 0 / 1**

Options

A. For labeling the component

B. For setting GPIO voltage

Expected

C. For accurate mechanical design

D. For choosing the programming language

FEEDBACK

Solution: Tolerances affect mechanical fit and must be considered when designing the PCB or housing.

Link: https://youtu.be/JeioUM7OGXQ?feature=shared&t=1020